CS Principles

Unit Journal Interactive Notebook

1: Digital Information

Getting Started: Directions

Print out copies of each page for each student. Each page has its own list of Materials Needed.

There are various Unit Journal activities that are associated with lessons throughout this unit of study.



Look for the *Unit Journal icon* in course slides in order to know when there is an associated activity in the Unit Journal.

Write down a question that has ____ possible answers, and brainstorm a device that can communicate the different responses using only classroom supplies

Materials Needed

Pencil or Pen
Scissors
Brainstorming materials

EXAMPLE

	2 answers	4 answers	8 answers
Question			
Device			

Interactive Notebook Directions

1. Cut out the question to the left and paste it into your notebook.

2. Divide a page in your notebook into 3 columns and 2 rows.

3. Label the rows *Question* and *Device*.

4. Label the columns *2 answers*, *4 answers*, and *8 answers*.

5. Following your teacher's directions for the activity, in each column, write down a question, then brainstorm a device.

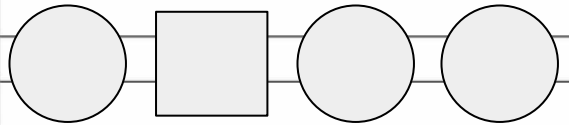
Materials Needed

Pencil or Pen
Tape or Glue

EXAMPLE

Notes

The main idea of today's lesson was...



Another group's rule:

Interactive Notebook Directions

1. Take notes, while answering the prompt
   - The main idea of today's lesson was...

Combine written notes with simple sketches or doodles to help you remember key concepts or ideas. This technique can make your notes more engaging and easier to review.

2. When you have completed the Circle Square Patterns activity, paste one of your four place value patterns on the page.

3. Write down an interesting rule that another group shared.

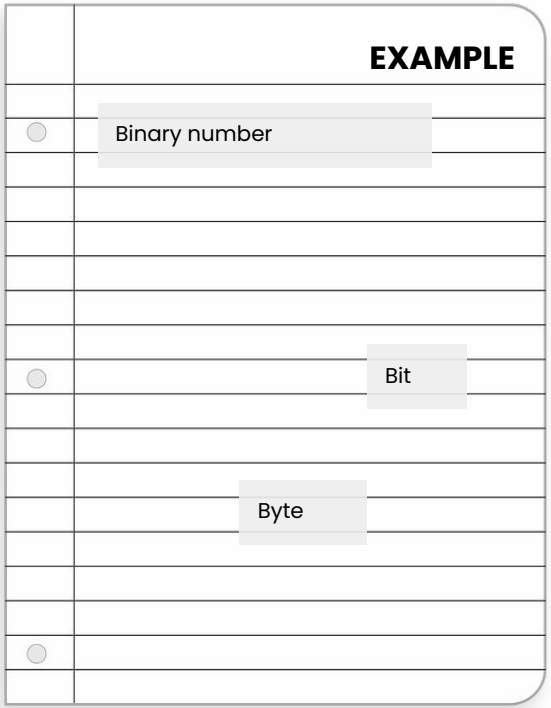
Materials Needed

Pencil or Pen
Scissors
Tape or Glue

Binary number

Bit

Byte



Interactive Notebook Directions

1. Cut out the vocabulary words to the left and paste it into your notebook.

2. Add definitions, images, and representations that are unique and meaningful to you.

Materials Needed

Pencil or Pen

EXAMPLE

Notes

Overflow and roundoff errors from today's lesson reminds me of...

Challenge question:

Interactive Notebook Directions

1. Take notes, while answering the prompt
   - Overflow and roundoff errors from today's lesson reminds me of...

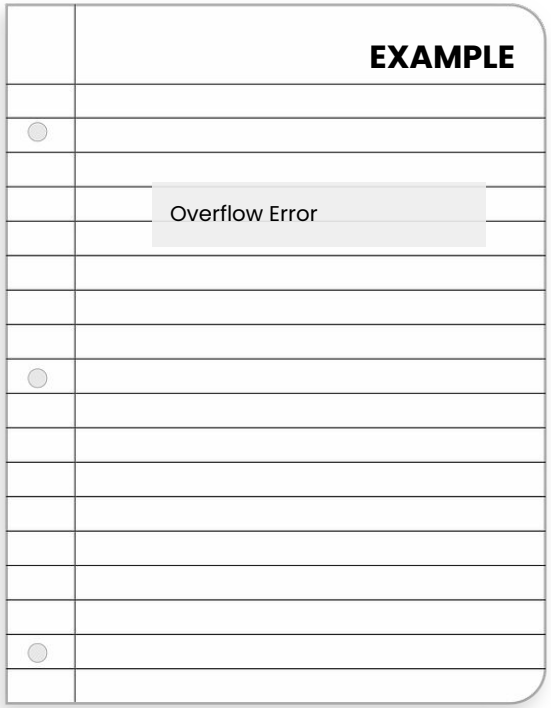
Combine written notes with simple sketches or doodles to help you remember key concepts or ideas. This technique can make your notes more engaging and easier to review.

2. Write responses to challenge questions in your journal!

Materials Needed

Pencil or Pen
Scissors
Tape or Glue

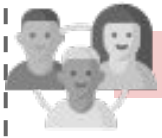
Overflow Error



Interactive Notebook Directions

1. Cut out the vocabulary words to the left and paste it into your notebook.

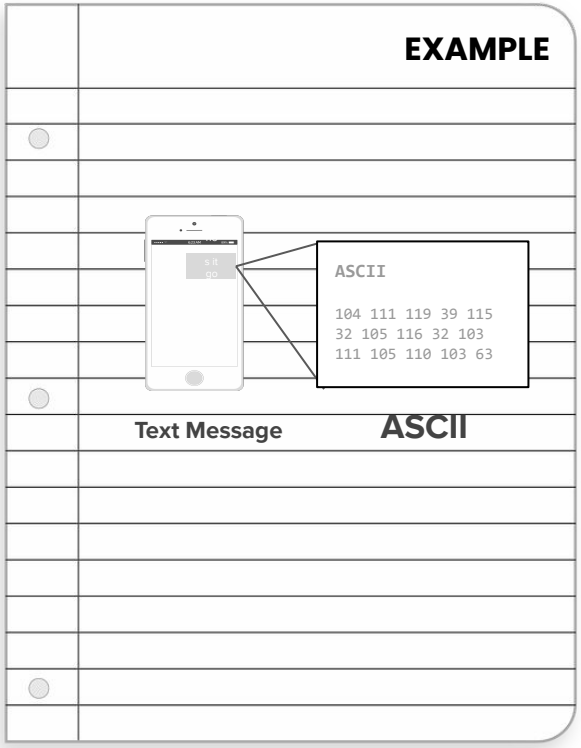
2. Add definitions, images, and representations that are unique and meaningful to you.



Describe / Diagram a non-computing example of an abstraction from your home, your neighborhood, your school, your culture, or your identity.

Materials Needed

Pencil or Pen
Scissors; Tape or Glue
Brainstorming materials



Interactive Notebook Directions

1. Cut out the question to the left and paste it into your notebook.

2. Use the brainstorming materials to create your abstraction. You can draw / sketch a diagram, write, doodle, or use any of the brainstorming materials to be creative!

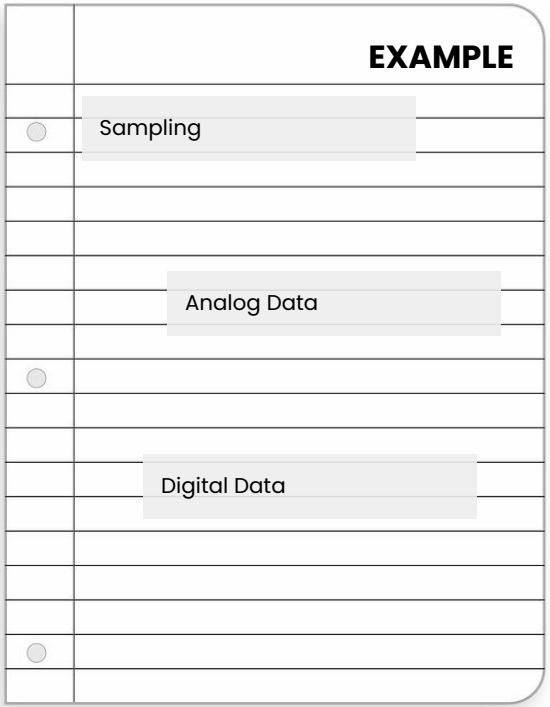
Materials Needed

Pencil or Pen
Scissors
Tape or Glue

Sampling

Analog Data

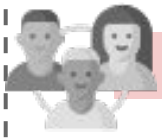
Digital Data



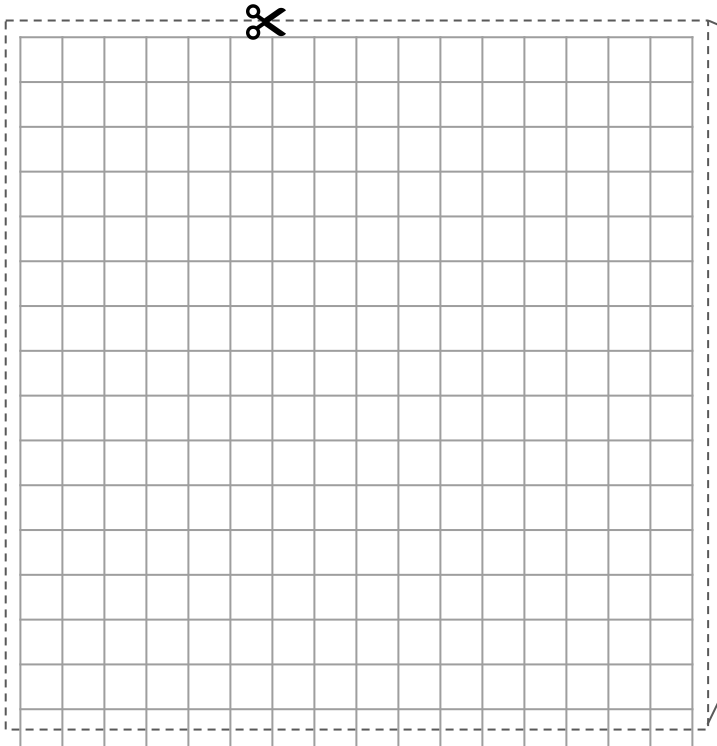
Interactive Notebook Directions

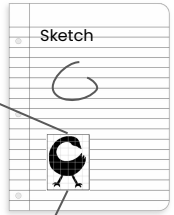
1. Cut out the vocabulary words to the left and paste it into your notebook.

2. Add definitions, images, and representations that are unique and meaningful to you.



Create a logo/symbol that is meaningful to you and which you might wear on the back of your cape if you were a superhero.





Materials Needed

Pencil or Pen
Scissors

Interactive Notebook Directions

1. Cut out the question to the left and paste it into your notebook.

2. Sketch a logo/symbol.

3. Cut out the grid and paste it into your notebook.

4. Use this grid to sample your original design and to create a pixellated version of it.

Materials Needed

Pencil or Pen

EXAMPLE

Notes

One question I had during the lesson was...

Interactive Notebook Directions

1. Take notes, while answering the prompt
   - One question I had during the lesson was...

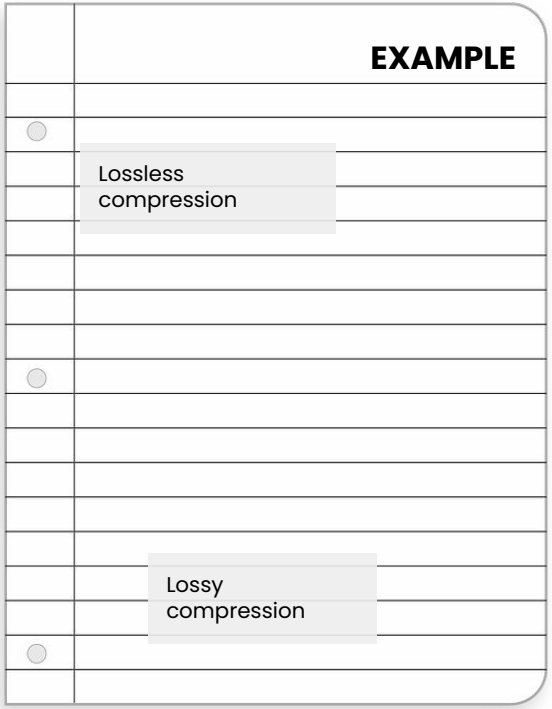
Combine written notes with simple sketches or doodles to help you remember key concepts or ideas. This technique can make your notes more engaging and easier to review.

Materials Needed

Pencil or Pen
Scissors
Tape or Glue

Lossless compression

Lossy compression



Interactive Notebook Directions

1. Cut out the vocabulary words to the left and paste it into your notebook.

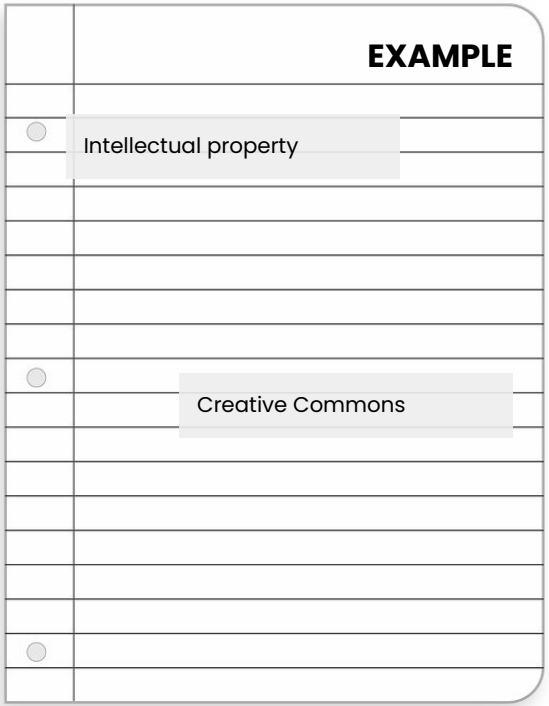
2. Add definitions, images, and representations that are unique and meaningful to you.

Materials Needed

Pencil or Pen
Scissors
Tape or Glue

Intellectual property

Creative Commons



Interactive Notebook Directions

1. Cut out the vocabulary words to the left and paste it into your notebook.

2. Add definitions, images, and representations that are unique and meaningful to you.

Materials Needed

Pencil or Pen
Scissors
Tape or Glue

Binary number	Lossy Compression
Intellectual Property	Byte
Overflow Error	Bit
Decimal Number	Creative Commons
Round-off Error	Lossless Compression
Sampling	Analog Data
Digital Data	

Interactive Notebook Directions

1. Cut out the vocabulary words to the left and paste it into your notebook.

2. Use lines and text to create connections between the various vocabulary words that you have been creating with your Vocabulary Canvas journal entries. Not all terms might connect. Sometimes words might have multiple connections that you can explore. A sample connection between bits and bytes is completed below.

